AP PRECALCULUS

Unit 1 • Worksheet 1.9-1.10A

Worksheet 1.9-1.10A • Rational Functions and Asymptotes

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2-x-6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Question 2

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2+x-2}{x^2-1}$?

(A) $\left(1, \frac{3}{2}\right)$

(B) $(-1, 0)$

(C) $(1, 0)$

(D) $(-2, 0)$

Question 3

NO CALCULATOR

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) ∞

(B) $-\infty$

(C) 2

(D) 0

Question 4 NO CALCULATOR

What happens on the graph of $r(x) = \dfrac{x-2}{(x-2)^2}$ at $x = 2$?

(A) a hole
(B) a zero
(C) a vertical asymptote
(D) nothing

Question 5 (Free response) NO CALCULATOR

For each function, find all vertical asymptotes and holes. For each asymptote, give both one-sided limits. (a) $r(x) = \dfrac{x-1}{(x+2)(x-3)}$ (b) $s(x) = \dfrac{x^3-4x}{x^2+2x}$ (c) $t(x) = \dfrac{x+1}{(x-2)^2}$